

SHRIDHAR PATHAK

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PROFESSIONAL SUMMARY

Senior Director of AI with extensive leadership in building large-scale AI platforms and multi-agent conversational systems. Leads AI strategy and engineering at Moderna, delivering enterprise GenAI solutions across multiple business units, and previously launched Alexa's first multi-agent conversational framework powering global voice experiences. Proven track record of scaling high-reliability, low-latency AI infrastructure and driving cross-functional teams to launch AI-powered products worldwide.

Seeking to apply deep technical and strategic expertise to accelerate AI innovation and business impact.

CORE LEADERSHIP & TECHNICAL EXPERTISE

Conversational Agents & Multi-Agent Systems, LLM Orchestration, Fine-Tuning Strategy, Applied Generative AI Infrastructure, AI Platform Architecture & Developer Experience, Cloud-Native ML Systems (AWS, Distributed Microservices), AI Governance & Enterprise Compliance, Engineering Org Design & Global Delivery, Cross-Org Strategy (Research, Product, Infra, Legal, External Partners), MLOps & Production ML Lifecycle.

PROFESSIONAL EXPERIENCE

Senior Director, AI & ML: Moderna | Cambridge, MA | 2023 – Present

- Defined and executed Moderna's generative AI roadmap for 20+ enterprise applications, including conversational agents, function calling, and LLM orchestration workflows for R&D, manufacturing, and clinical operations.
- Built and led cross-functional AI/ML engineering teams (35+ engineers) delivering production LLM-based applications integrated with cloud infrastructure (AWS) using Python and MLOps pipelines (30+).
- Architected multi-agent conversational systems leveraging fine-tuning, distillation, and RAG (retrieval-augmented generation) to automate knowledge work across regulated enterprise workflows for all document systems at Moderna.
- Partnered with C-suite and business unit leaders to translate AI technical capabilities into measurable business outcomes, accelerating go-to-market strategy and stakeholder buy-in across pharmaceutical R&D, quality, and regulatory teams. Established strategic partnerships with OpenAI, AWS, and cloud AI vendors to integrate cutting-edge generative AI technologies into enterprise platforms.

Amazon: Senior Manager & Tech Lead, Alexa | Seattle, WA | 2022 – 2023

Senior engineering leader responsible for large-scale conversational AI and multi-agent orchestration frameworks powering Alexa globally, managing 35+ direct report and 110+ partner team reports spread across 8+ countries. Key Contributions:

- Architected and launched Alexa's first multi-agent conversational switching framework. Enabled seamless orchestration across celebrity personalities (Melissa McCarthy, Shaquille O'Neal, Amitabh Bachchan) and partner agents (Disney). Designed contextual aggregation and routing services handling millions of low-latency voice requests daily. Built scalable architecture supporting dynamic agent selection and personality switching.
- Built and scaled multi-disciplinary teams (35+ engineers across regions). Drove cross-org alignment across Alexa AI, retail, device, and international engineering teams. Set architectural standards for reliability and developer extensibility. Impact: Delivered production conversational AI systems used by millions of customers globally, forming the backbone of agentic consumer AI experiences.

Amazon: Principal Technical Program Manager, Alexa | 2020 – 2022

Pioneered the first multi-agent conversational AI framework, enabling monetization and engagement measurement for celebrity personalities. Key Contributions:

- o Launched Alexa Voice Shopping across US, India, Japan, Canada, France, Italy, Spain, Mexico, Ireland, and Australia. Delivered international AI product rollouts requiring localization, regulatory adaptation, and infrastructure scaling.
- o Led integration of speech recognition, NLP, contextual understanding, and response orchestration. Defined APIs and service endpoints enabling ML-powered interactions in the Amazon India retail app (millions of DAU). Balanced personalization, privacy, latency, and reliability at global scale.

Senior Technical Program Manager, Alexa India | 2018 – 2020

Founding technical member of the Alexa India Retail team, building the organization from scratch to 16+ members and integrating Voice AI into the Amazon Retail App. Key highlights:

- o Led the integration of Alexa into the Amazon India retail app, architecting the APIs and ML endpoints that allowed millions of users to shop via voice, a massive shift in consumer behavior measurement.
- o Built and mentored a high-performing team of 16 engineers and analysts, establishing the culture and best practices for retail analytics in a new market.

Manager & Tech Lead, Amazon Retail (Southeast Asia Launch) | 2016 – 2018

Core technical leader for Amazon's first retail entry into Southeast Asia, responsible for architecting the region's marketing, routing, and operational technology stack from the ground up. Key highlights:

- o Reduced the standard 3-year country launch timeline to 9 months for Amazon's first Southeast Asia retail entry by decoupling legacy US-centric systems to create a lightweight, region-specific operating model. Acted as the single-threaded technical owner, orchestrating roadmap alignment across 26 independent teams and seven global organizations to meet flagship leadership goals.
- o Architected the region's first end-to-end digital marketing and last-mile logistics stack, enabling the launch of proprietary two-hour delivery services in Singapore. Engineered a scalable marketing engine for automated customer lifecycle management and built a logistics network that fulfilled 100% of regional orders via proprietary systems.

Amazon: Software Development Engineer: 2012 – 2016

Built distributed services for Amazon Register (POS platform). Designed onboarding, payments, and integration systems for merchants. Early career foundation in high-scale, fault-tolerant distributed systems.

Tata Institute of Fundamental Research (TIFR) | Junior Research Fellow|2009 – 2010

Designed detector-monitoring algorithms under CERN experimental frameworks. Built scientific computing systems using C++ and ROOT.

Other

Noesis: Enterprise Agent Workforce Platform | Tech Advisor-2025 – Present

Conceived and architected a workforce management layer for enterprise-scale AI agents, positioning AI systems as a managed "digital workforce." Designed Policy-as-Code and runtime supervision frameworks enabling real-time monitoring, intervention, and compliance for LLM-driven multi-agent systems.

EDUCATION

MIT Sloan School of Management: Executive MBA (2026), Cross registration at Harvard Business School.

University of Texas at Dallas: Master of Science, Computer Science (2012)

Pune Institute of Computer Technology: Bachelor of Engineering, Computer Science (2009)

AWARDS & LEADERSHIP RECOGNITION

Stanford Seed Consultant — Stanford Graduate School of Business

Selected strategic advisor within the Seed Transformation Network, partnering with high-growth global founders to scale operations, strengthen AI-enabled business models, and drive sustainable expansion.

Engineering Leadership Development — Rice University

Advanced executive training in leading high-performing technical teams, organizational alignment, and cross-functional relationship management.

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow — Government of India

Nationally recognized Young Scientist Scholar for academic and research excellence in engineering.